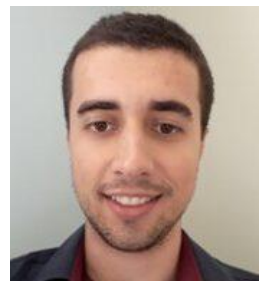


Mooring System Analysis and Fragility Curve Estimation: The Economic Impact of Low Probability High Impact Events on Ocean Current Devices

Anderson Rodrigo de Queiroz, Mohammed Gabr

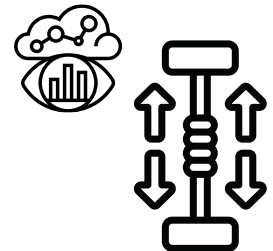
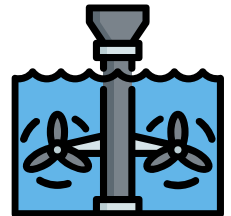
Neda Jamaledin, Victor Faria

March 20th, 2023

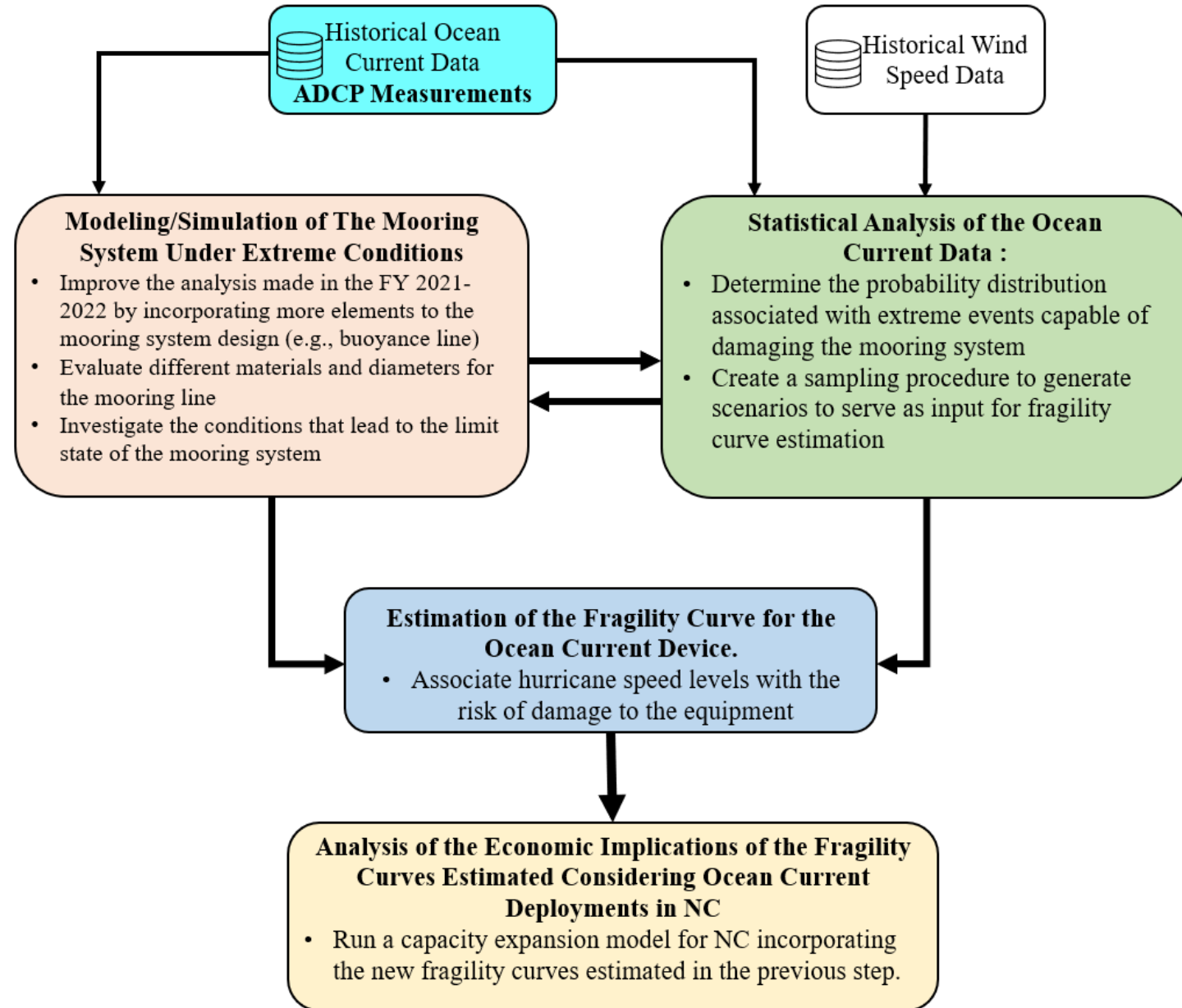


Introduction

- The US government has recently announced a 30 GW target for offshore wind energy deployment by 2030 and 110 GW by 2050 (WH, 2022). More locally, NC has defined as its objective to **reduce by 70%** its CO2 emissions by 2030 and reach **carbon neutrality by 2050** (GANC, 2021)
- In the future marine hydrokinetic devices can contribute to a more **diversified** offshore renewable energy portfolio, helping to **reduce energy variability and improve system security** (Li et al, 2017; Faria et al, 2022). However, the **vulnerability** of these technologies to extreme events such as hurricanes has not been thoroughly studied
- This research proposes the use of **data analytics** and mechanical model simulations using the **Ansys-AQWA software** to perform mooring system analysis for ocean current devices, and the integration of those analyses in the construction of **fragility curves** and **risk assessment** of ocean current deployments
- Our fragility curves associate **hurricane speed levels with the risk of damage to the equipment** and were incorporated into a capacity expansion model for the NC energy system to assess how the economics of this device affects its development in the NC power grid



Project Flow Diagram



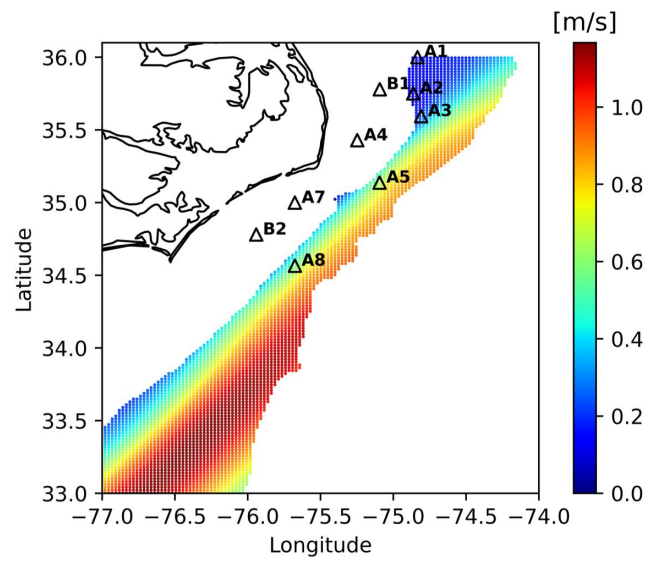


Figure 1. ADCP Locations and Average Daily Ocean Current Speed from 2009 to 2013 considering the MABSAB model (Gong, et al. 2015)

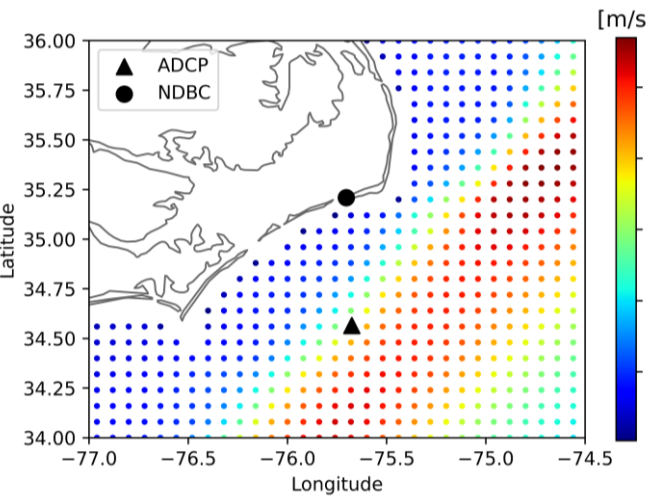


Figure 2: ADCP A8 and NDBC Station Together with the Spatial Distribution of the Ocean Current Resource at 50m Depth (HYCOM 1994-2021)

Data

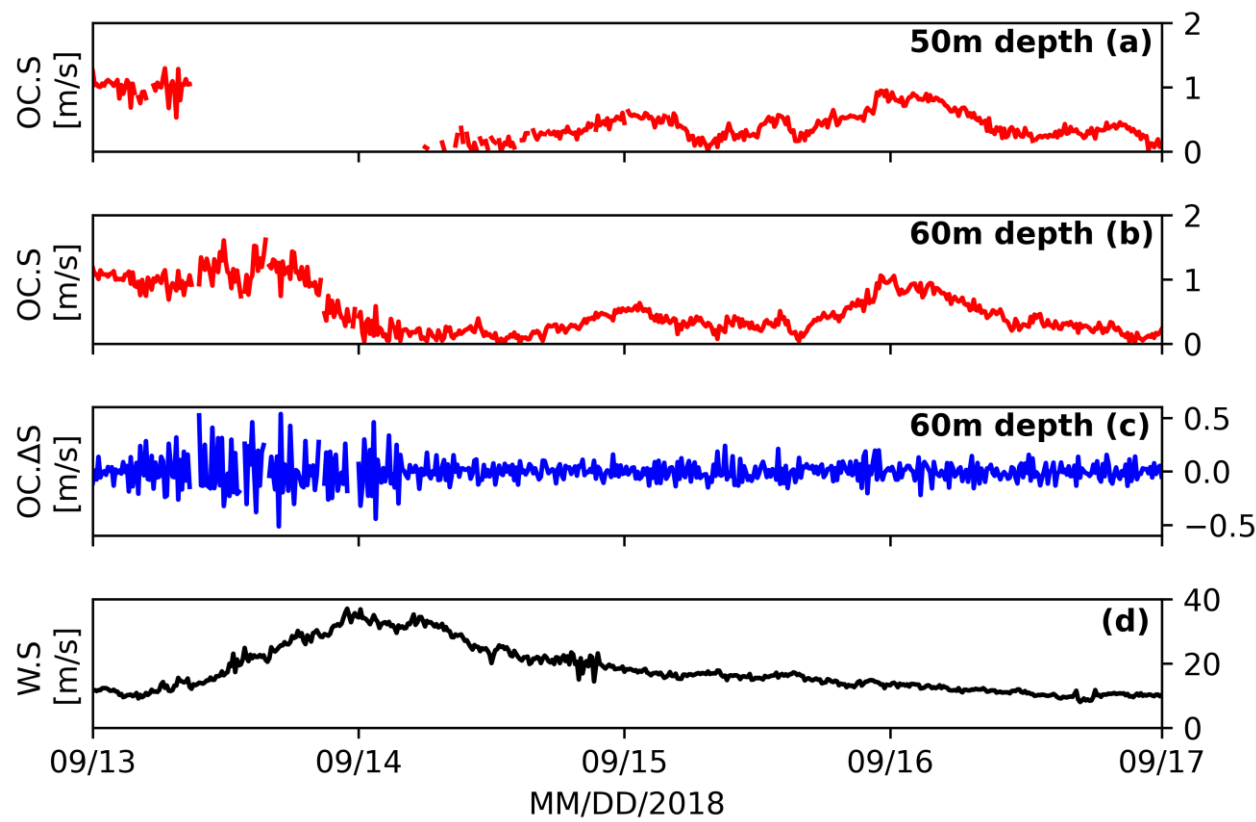


Figure 3. Ocean and Wind Condition During Hurricane Florence. a) Ocean Current Speed at 50[m] b) Ocean Current Speed at 60[m] depth; c) Ocean Current Speed Variation at 60[m] depth; d) Wind Speed

Statistical Analysis

(100-year Recurrence Curve)

- **100 Year Recurrence Curve:** What are the most extreme conditions expected to repeat every 100 years?
- In general, recurrence curves are constructed by **fitting a probability distribution** to a set of observations and **finding through this distribution** the set of extreme events that are expected to repeat only every a certain period of time

Model I: Joint Probability Distribution ($S, \Delta S$)

$$(S, \Delta S) \sim \text{Normal}(\mu, \Sigma) \quad (1) \quad \sqrt{\Sigma_{22}} \sim \text{Gamma}(a, a) \quad (5)$$

$$\mu_s = \beta_{0,s} + \beta_{1,s}D \quad (2) \quad \Sigma_{12} = \sqrt{\Sigma_{11}\Sigma_{22}} \times \rho \quad (6)$$

$$\mu_{\Delta s} = \beta_{0,\Delta s} + \beta_{1,\Delta s}D \quad (3) \quad \rho \sim \text{Uniform}(-1,1) \quad (7)$$

$$\sqrt{\Sigma_{11}} \sim \text{Gamma}(a, a) \quad (4) \quad \beta \sim \text{Normal}(0, 1/a) \quad (8)$$

Statistics of the Intensity of the Extreme Event

Model II: Probability Distribution of the Number of Extreme Events

$$Y \sim \text{Poisson}(\lambda \Delta T) \quad (9) \quad \lambda \sim \text{Gamma}(a, a) \quad (10)$$

$$\lambda | Y \sim \text{Gamma}(a + Y, a + \Delta T) \quad (11)$$

Statistics of the Frequency of the Extreme Events

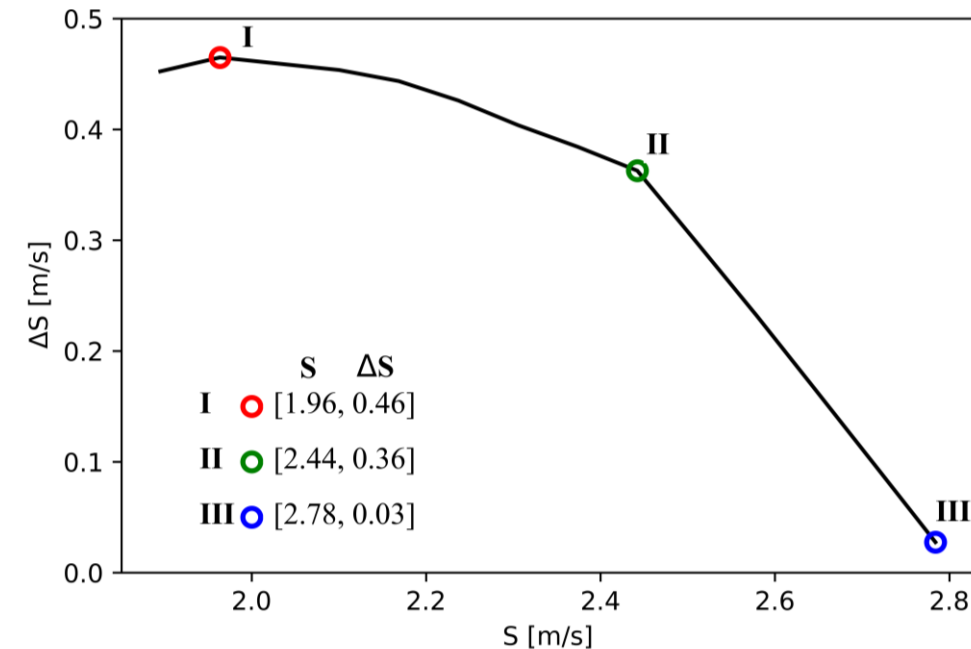


Figure 4. 100-Year Recurrence Curve: Worst Events Every 100-Years

Mooring System Analysis

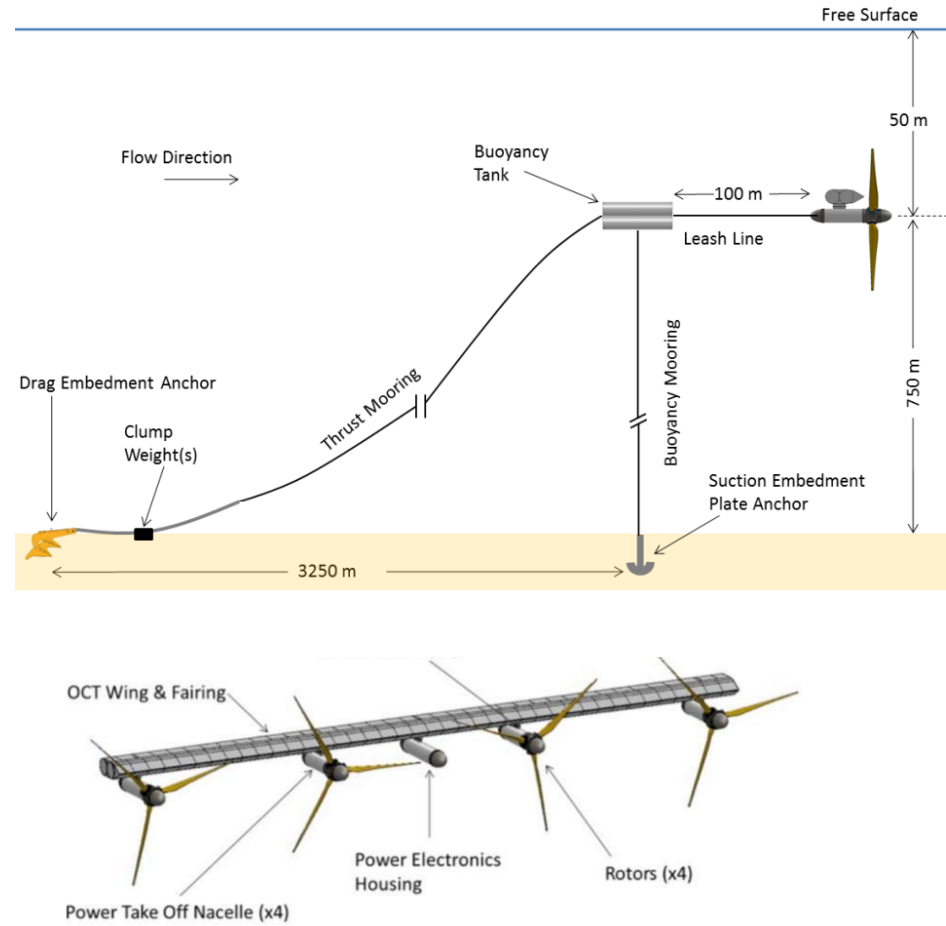


Figure 5. RM4 device configuration and dimension, Neary et al (2014)

Mooring System Analysis (cont.)

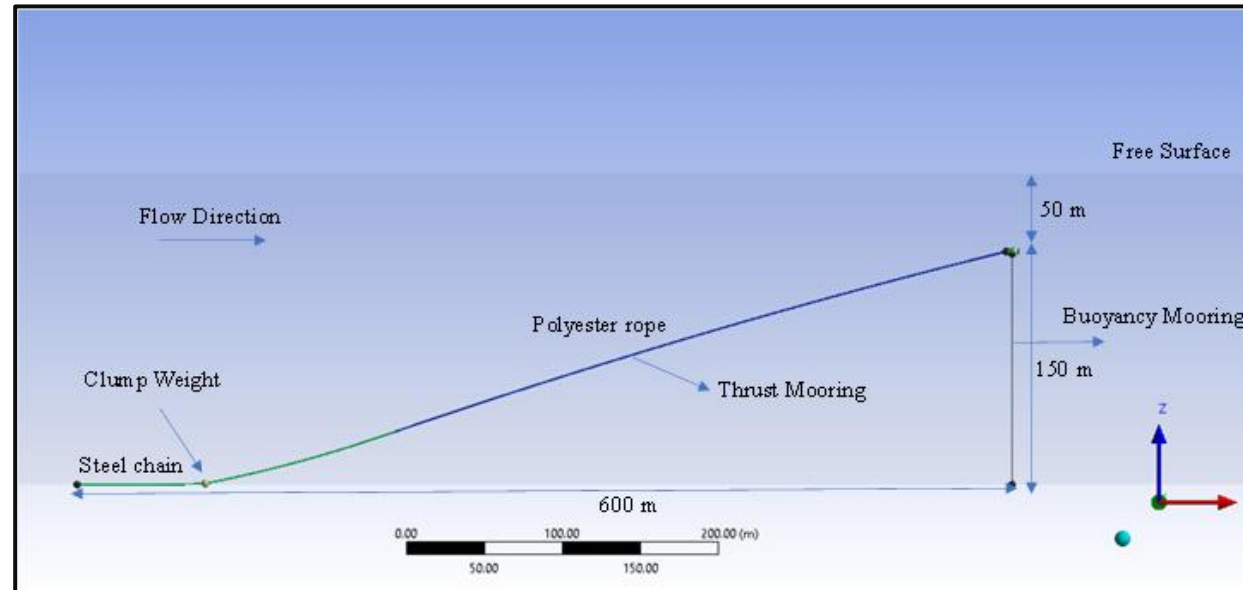


Figure 6. Schematic of the adopted model using AQWA

Table 1. Utilized parameters in analyses

Parameter	Unit	Value
Depth of water	m	200
Buoyancy tank volume	m ³	452
Buoyancy tank mass	ton	150
Thrust mooring line length	m	610

Design Criteria

Ultimate Limit State (ULS) design criteria for dynamic analysis according to Det Norske Veritas, DNV's Offshore Requirements for Position Moorings:

$$T_{mean}\gamma_{mean} + T_{dyn}\gamma_{dyn} \leq 0.95CBS$$

In case of consequence class 3, a partial safety factor (γ_{dyn}) of 1.5 for dynamic tension and a partial safety factor (γ_{mean}) equal to 1.1 for mean equilibrium tension need to be considered.

Table 3. Consequence class associated possible outcomes of a mooring system failure per IEC TS 62600-10

Consequence class	Possible outcomes
3	Loss of human life, significant damages
2	Serious human injury and damages
1	Minimal human injury and damages

Mooring lines tension – Extreme events

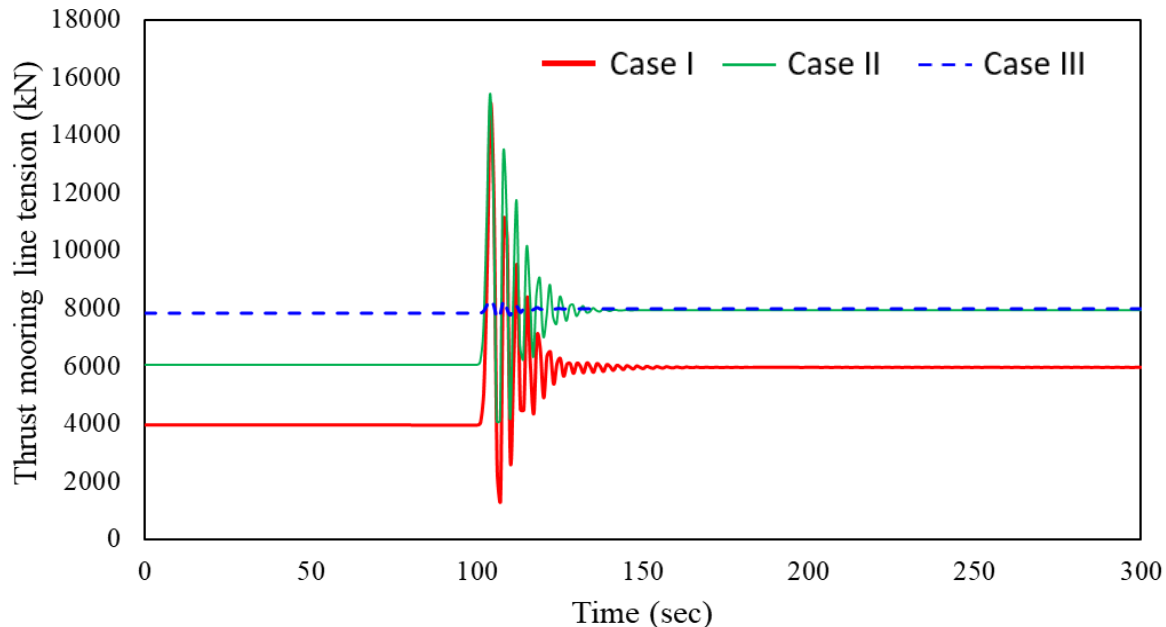


Figure 7. Thrust mooring line (with $D = 280\text{mm}$) tension versus time in different scenarios.

By applying partial safety factors, it was concluded that a thrust mooring line with diameter of 280mm can withstand the 100-year extreme ocean current conditions

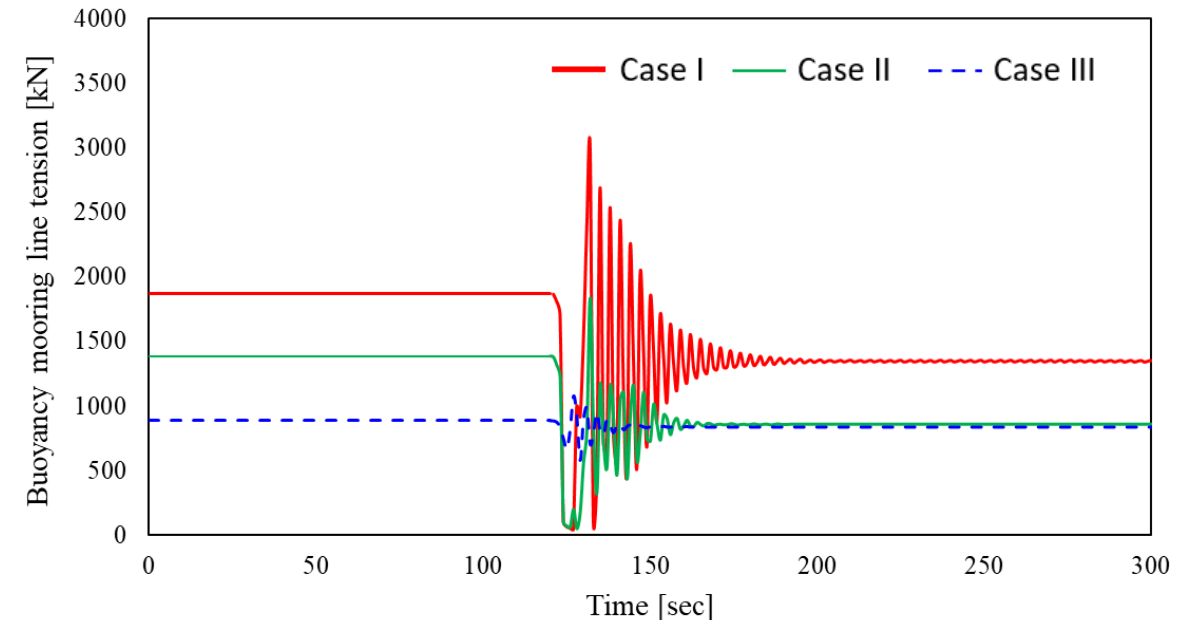
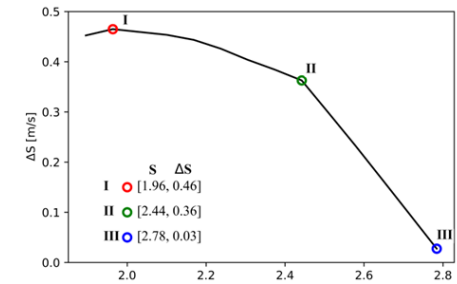


Figure 8. Buoyancy Mooring line (with $D = 250\text{mm}$) tension versus time in different scenarios.

The minimum required buoyancy line diameter was determined to be 250mm, by restricting the static vertical displacement of the buoyancy tank to 1m



Different mooring configurations

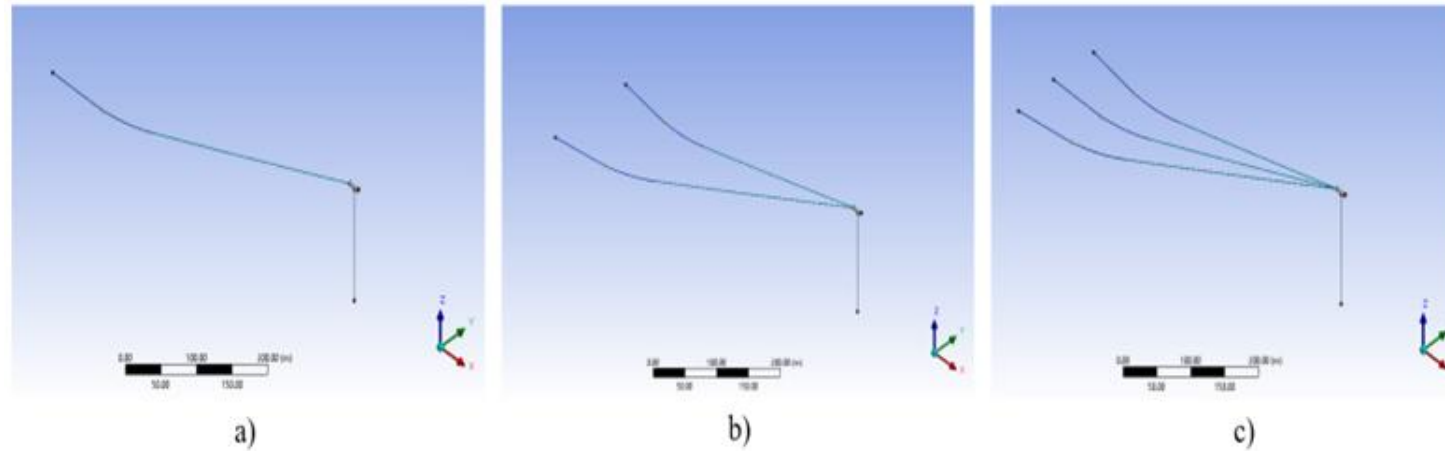


Figure 9. Schematic 3D view of a) configuration 1 b) configuration 2, and c) configuration 3

Table 4. Thrust Mooring Line Configurations.

Polyester thrust mooring line(s)	Number of mooring lines	Diameter (mm)		CBS (kN)	
		chain	rope	chain ^c	rope ^d
Configuration 1	1	180	280	23880	23000
Configuration 2 ^a	2	124	215	13048	13000
Configuration 3 ^b	3	89	170	7274	7225

^a 2 mooring lines with 15.0- degree separations

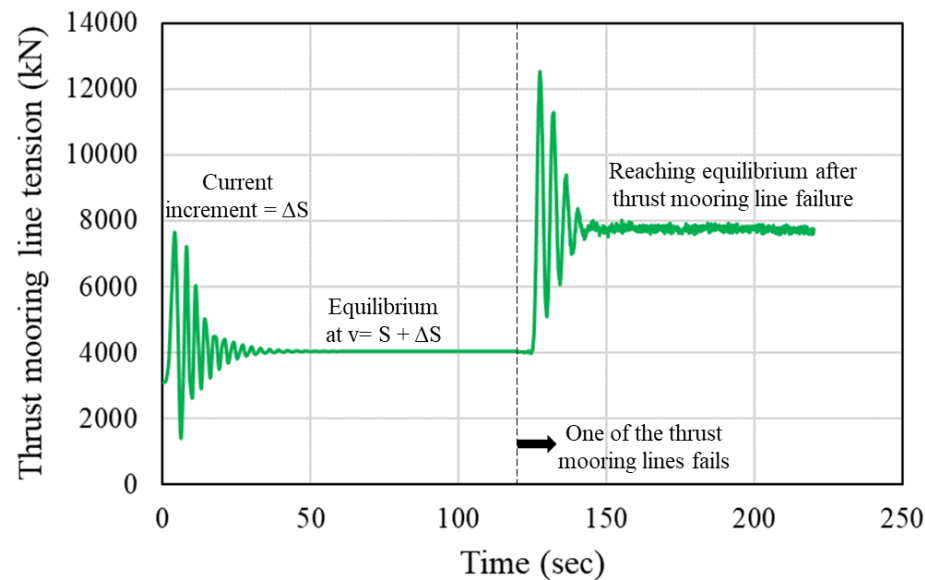
^c Aceton group Ltd., engineering, mooring, and foundations

^b 3 mooring lines with 7.5- degree separations

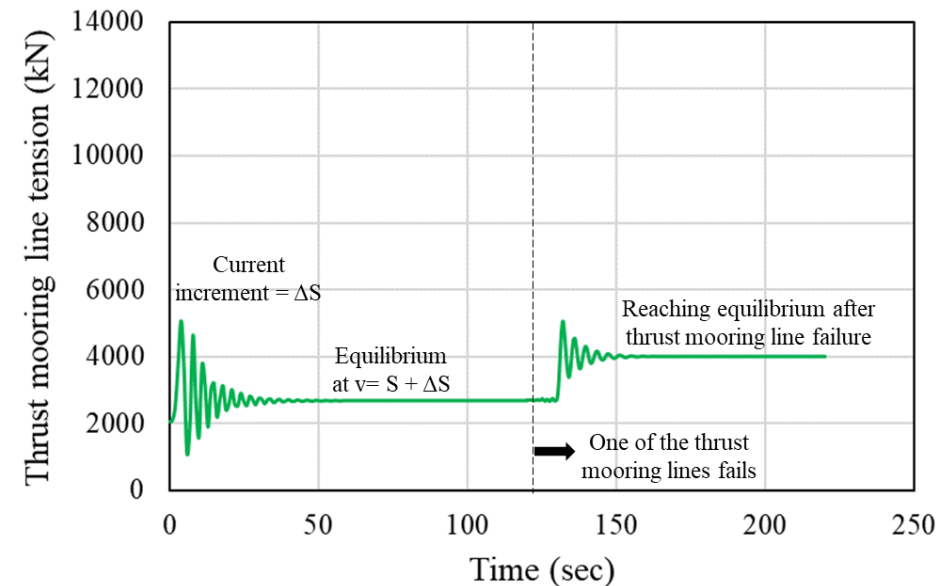
^d Mooring Ropes for Offshore Applications, Parker Scanrope AS

Different mooring configurations

The design of the thrust mooring line in configuration 2 and 3 is in a way that they meet the ULS design criteria, and if one of the thrust mooring lines fail, the other one(s) would not fail subsequently (i.e., tension will not exceed the CBS)



a) Configuration 2 (CBS: 13.0 MN)



b) Configuration 3 (CBS: 7.2 MN)

Figure 10. Thrust mooring line tension versus time under an extreme event (case 2) in a) configuration 2, b) configuration 3

Fragility Curve Estimate

Model III: Joint Probability Distribution ($S, \Delta S$) Under Extreme Winds

$$(S, \Delta S) \sim \text{Normal}(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \quad (12)$$

$$\mu_s = \beta_{0,s} + \beta_{1,s}D + \beta_{2,s}WS \quad (13)$$

$$\mu_{\Delta s} = \beta_{0,\Delta s} + \beta_{1,\Delta s}D + \beta_{2,\Delta s}WS \quad (14)$$

$$\sqrt{\Sigma_{11}} \sim \text{Gamma}(a, a) \quad (15)$$

$$\sqrt{\Sigma_{22}} \sim \text{Gamma}(a, a) \quad (16)$$

$$\Sigma_{12} = \sqrt{\Sigma_{11}\Sigma_{22}} \times \rho \quad (17)$$

$$\rho \sim \text{Uniform}(-1, 1) \quad (18)$$

$$\beta \sim \text{Normal}(0, 1/a) \quad (19)$$

PDF of $\beta_{2,\Delta s}$

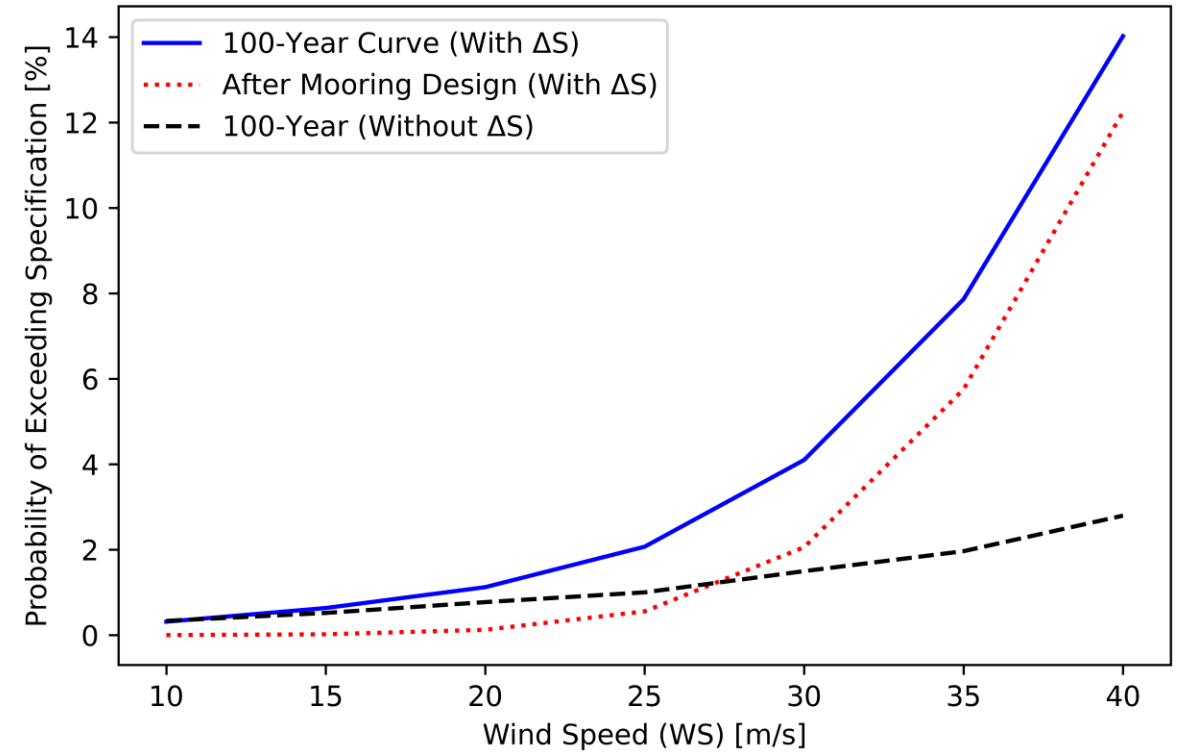
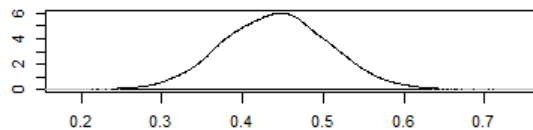


Figure 11. Fragility Curve of the Ocean Current Device Mooring System, Considering Different Maximum Operating Limits

Capacity Expansion Model



Temoa: An open-source framework for conducting energy system analysis.

<https://github.com/TemoaProject/temoa>

Figure 12. Scenario Tree (3 scenarios per stage, 5 stages). 243 Realizations

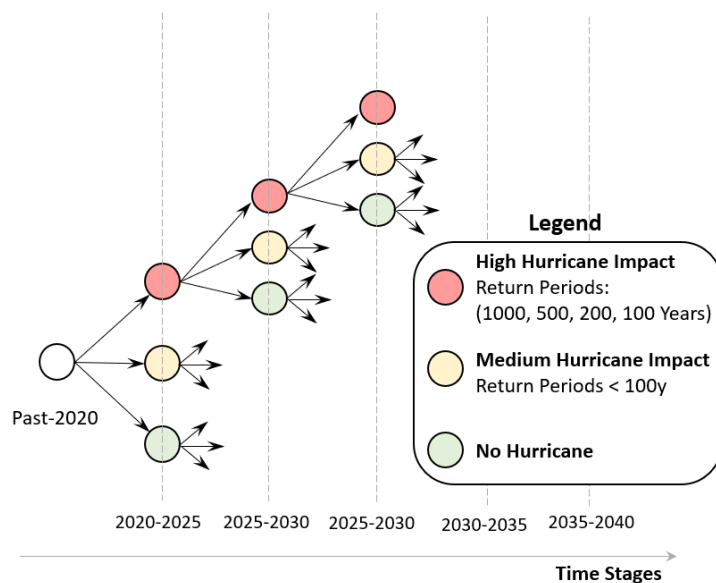
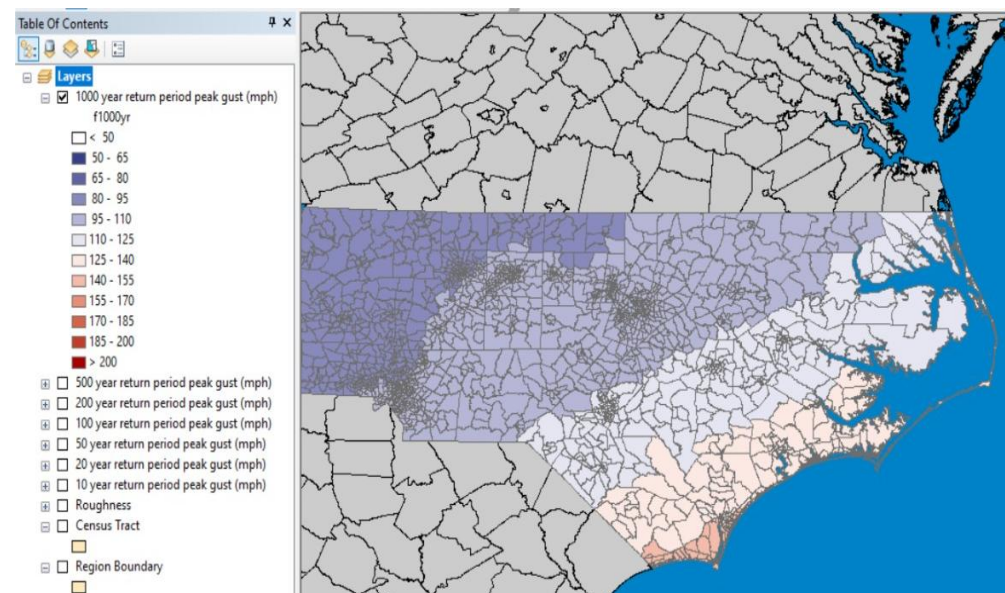
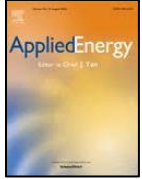


Figure 13. Hurricane Speed Data (FEMA - HAZUS)



This is a **stochastic optimization** model where **each technology has its own fragility**, and the expected technology **damage given a hurricane scenario** is computed using these curves and the statistics of hurricane speed from FEMA

Contributions



de Faria, V.A.D., de Queiroz, A.R., DeCarolis, J.F., (2022) Optimizing Offshore Renewable Portfolios Under Resource Variability, *Applied Energy*

de Faria, V.A.D., de Queiroz, A.R., DeCarolis, J.F., (2023) Scenario Generation and Risk-averse Stochastic Portfolio Optimization Applied to Offshore Renewable Energy Technologies, *Energy*



de Faria, V.A.D., Jamaledin, N., de Queiroz, A.R., Gabr, M., (2023) Assessing the Risk of Hurricane Damage to Marine Hydrokinetic Devices, *Ocean Engineering* (*under review*)

Planned Future Work

1. Continue investigating other mooring system configurations
2. Build fragility curves for the new mooring designs
3. Explore in more depth the capacity expansion model

de Queiroz, A.R., Gabr, M., Husain, I., Bonner, G., Elvin, G., Dubbs, L., Muglia, M., A Data-driven Technical, Economic, and Socio-Environmental Feasibility Assessment of the Atlantic Current Energy Site (ACES), DE-FOA-0002793, *February of 2023* (*under review*)

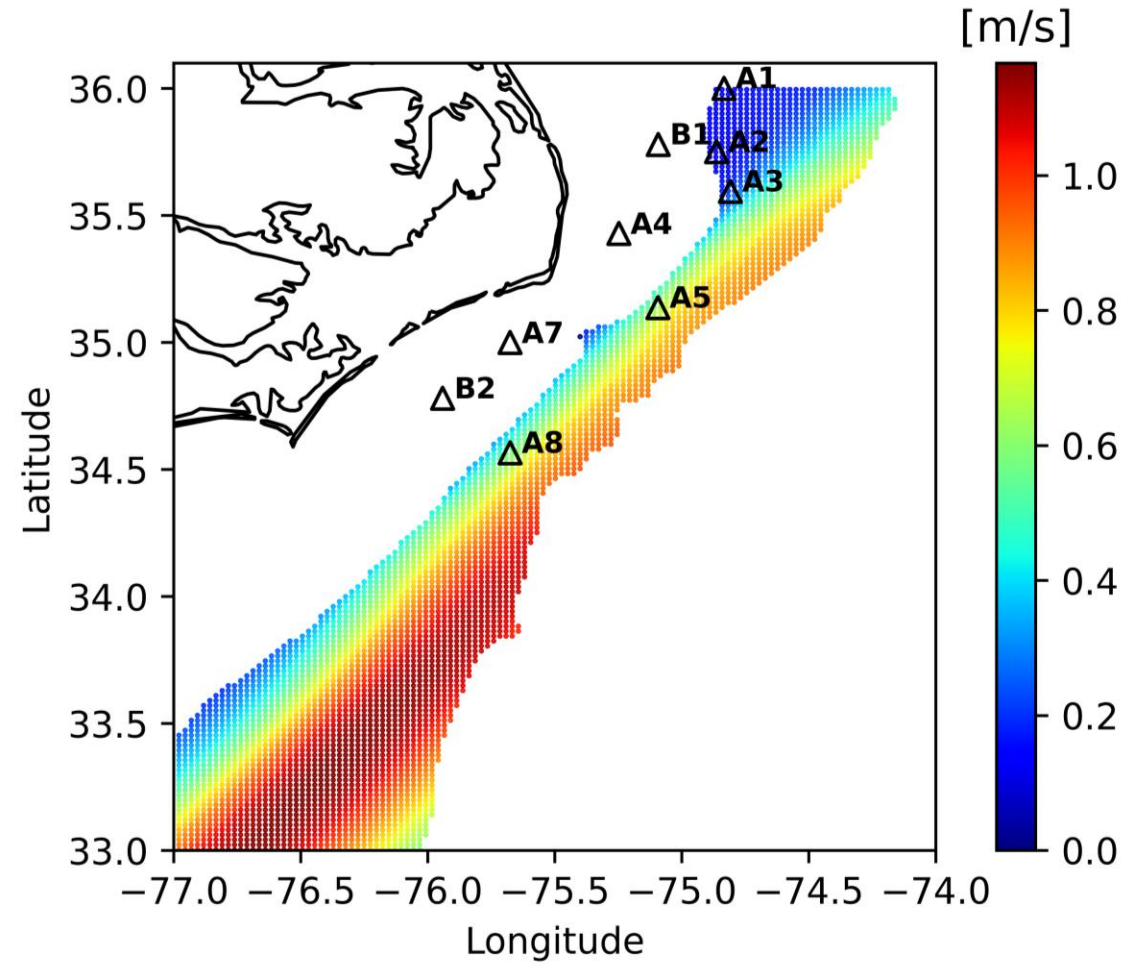


Thank You!



Data

Figure 1. ADCP Locations and Average Daily Ocean Current Speed from 2009 to 2013 considering the MABSAB model (Gong, et al. 2015)



Data

Figure 1. ADCP Locations and Average Daily Ocean Current Speed from 2009 to 2013 considering the MABSAB model (Gong, et al. 2015)

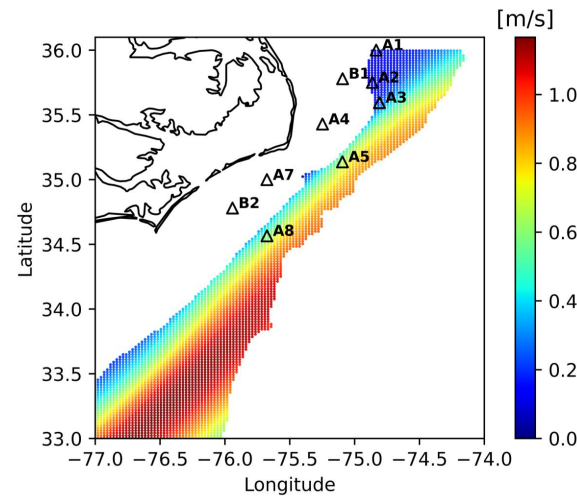
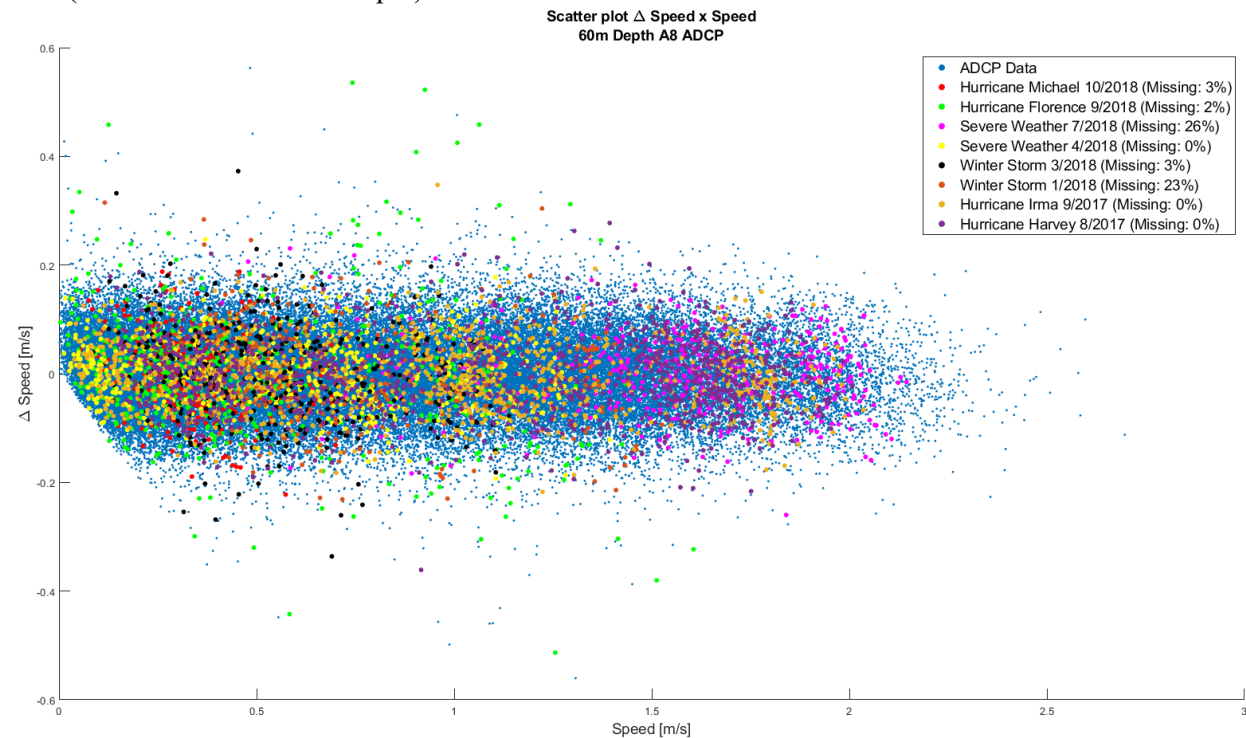


Figure 2. Scatter Plot of Current Speed and Current Speed Difference (Δ_s) at Consecutive Time Steps for ADCP A8 (April-2017 to November-2018), and Corresponding Weather Related Events. (Measurements at 60m Depth).



Thrust Force

Thrust force is modeled by applying a point force of the same value on the buoyancy tank towards the x-direction. The thrust force (T_h) is computed as:

$$T_h = 0.5 \rho_w A_R C_T U^2 = 0.125 \pi \rho_w C_T D^2 U^2$$

$$\left\{ \begin{array}{l} \rho_w \text{ is the water density: } 1025 \text{ kg/m}^3, \\ A_R \text{ is the rotor blades' swept area: } 855 \text{ m}^2 \\ C_T \text{ is thrust coefficient: } 0.55, \\ D \text{ is the rotor diameter: } 33 \text{ m}, \\ \text{and } U \text{ is current velocity, in m/s.} \end{array} \right.$$

Table 2. Equivalent thrust load produced by the turbine.

Current velocity (m/s)	Thrust force (kN)
0.5	240
1.5	2158
2.5	5995
2.8	7520

Statistical Analysis (100-year Recurrence Curve)

- **100 Year Recurrence Curve:** What are the most extreme conditions expected to repeat every 100 years?
- In general, recurrence curves are constructed by **fitting a probability distribution** to a set of observations and **finding through this distribution** the set of extreme events that are expected to repeat only every a certain period of time

Model I: Joint Probability Distribution ($S, \Delta S$)

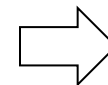
$$(S, \Delta S) \sim \text{Normal}(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \quad (1) \quad \sqrt{\Sigma_{22}} \sim \text{Gamma}(a, a) \quad (5)$$

$$\mu_s = \beta_{0,s} + \beta_{1,s}D \quad (2) \quad \Sigma_{12} = \sqrt{\Sigma_{11}\Sigma_{22}} \times \rho \quad (6)$$

$$\mu_{\Delta s} = \beta_{0,\Delta s} + \beta_{1,\Delta s}D \quad (3) \quad \rho \sim \text{Uniform}(-1,1) \quad (7)$$

$$\sqrt{\Sigma_{11}} \sim \text{Gamma}(a, a) \quad (4) \quad \beta \sim \text{Normal}(0, 1/a) \quad (8)$$

*a=0.005

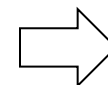


**Statistics of the Intensity
of the Extreme Event**

Model II: Probability Distribution of the Number of Extreme Events

$$Y \sim \text{Poisson}(\lambda \Delta T) \quad (9) \quad \lambda \sim \text{Gamma}(a, a) \quad (10)$$

$$\lambda | Y \sim \text{Gamma}(a + Y, a + \Delta T) \quad (11)$$



**Statistics of the Frequency
of the Extreme Events**

Equilibrium steady state thrust and buoyancy mooring line tensions

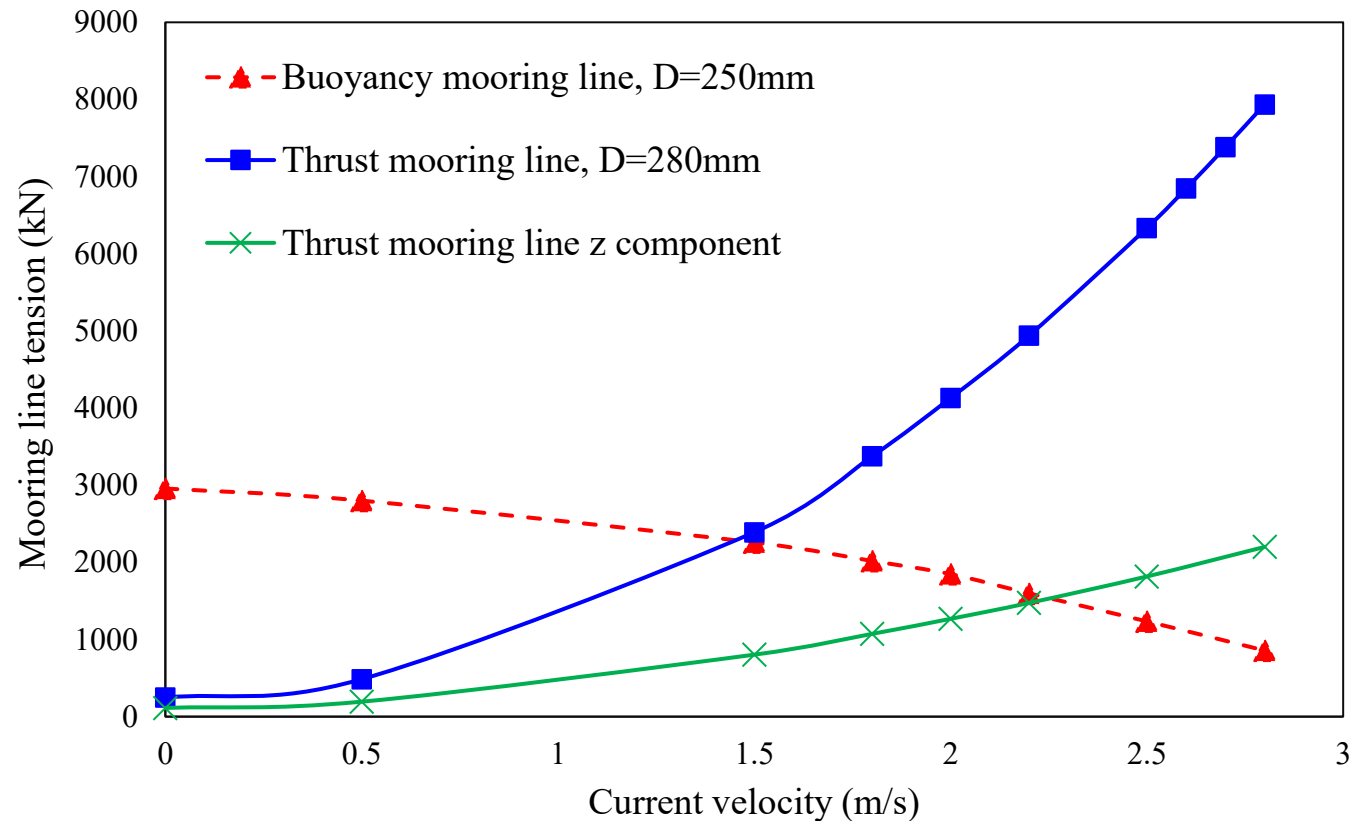
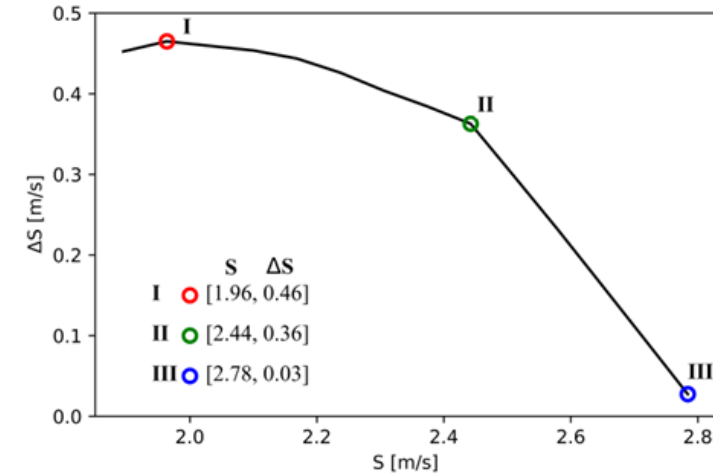
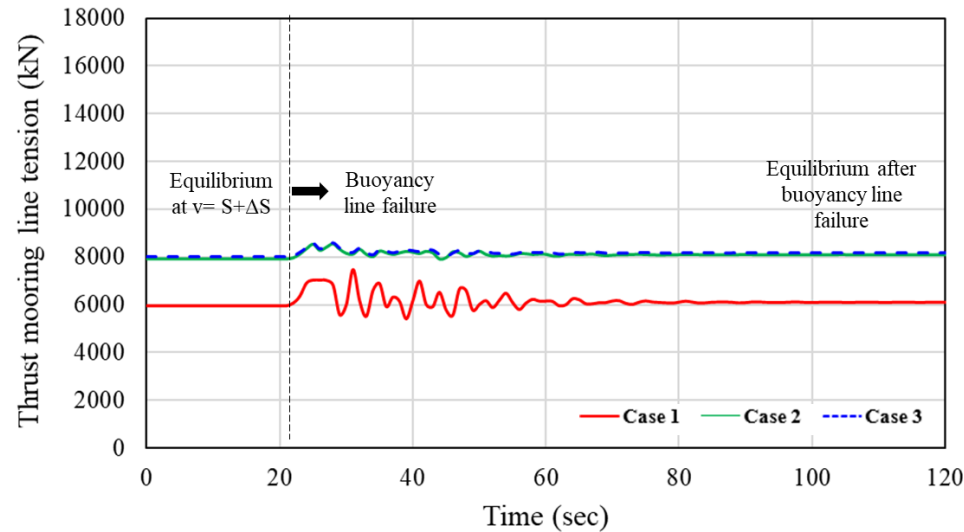


Figure 5. Tension forces in the thrust and buoyancy mooring lines as a function of the current velocity at the static equilibrium state.

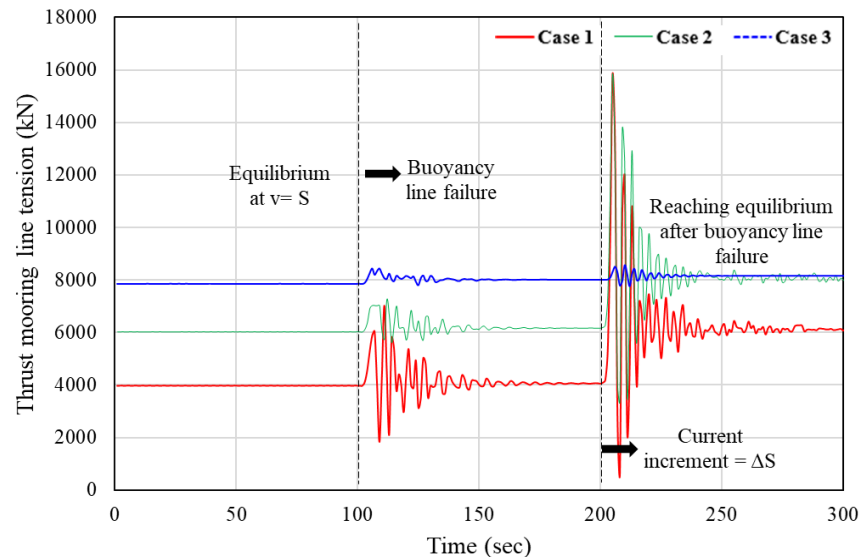
Buoyancy mooring line failure



Thrust mooring line: Polyester

$D = 280\text{mm}$

$CBS = 23000\text{ kN}$



In the design of thrust mooring line, the buoyancy mooring line failure and its impact on the thrust mooring line tension is considered.

Table 5. Properties for mooring line material (Baltrop 1998).

Mooring line material	CBS (N)		UTS (N/mm ²)	w (N/m)	EA (N)	CD	CM
Chain:							
OQR	c(44- 0.88D)D ²	<u>c</u> 21.1	641	0.1875 D ²	90000D ²	2.6- 3.0	-2.0
3		22.3	690			2.4	
3S		24.9	770				
4		27.4	860				
Wire rope:							
Six strands	525d ² -600d ²		1770- 1860	0.034d ²	45000d ²	1.5- 1.8	-2.0
Spiral strand	900d ²		1570	0.043d ²	90000d ²		
Fiber rope:							
Polyester	250d ²			0.0067d ²	5000d ² to 13000d ²		
Aramid	450d ²			0.00565d ²	15000d ² to 52000d ²		

MoorLine Polyester Mooring Ropes for Offshore Applications

Parker Scanrope AS

Metric table

MBL [TF]	MBL [kN]	Diameter [mm]	Mass in Air [kg/m]	Mass in Water [kg/m]	Maximum Length [m]
500	4905	138	12.2	3.1	8197
600	5886	149	14.3	3.7	6993
700	6867	160	16.4	4.2	6098
800	7848	169	18.4	4.7	5435
900	8829	178	20.4	5.3	4900
1000	9810	187	22.5	5.8	4450
1100	10791	195	24.4	6.3	4095
1200	11772	203	26.5	6.8	3776
1300	12753	213	29.4	7.6	3397
1400	13734	219	31.3	8.1	3195
1500	14715	226	33.4	8.6	2996
1600	15696	234	35.6	9.2	2809
1700	16677	240	37.5	9.7	2665
1800	17658	248	39.5	10.2	2532
1900	18639	254	42.5	10.9	2353
2000	19620	260	44.6	11.5	2242
2100	20601	266	46.5	12.0	2151
2200	21582	271	48.5	12.5	2062
2300	22563	277	50.7	13.0	1972
2400	23544	282	52.7	13.6	1898
2500	24525	290	55.8	14.4	1792
2600	25506	295	57.9	14.9	1727
2700	26487	300	59.9	15.4	1669
2800	27468	305	61.9	15.9	1616
2900	28449	310	64.0	16.5	1563
3000	29430	315	66.0	17.0	1515

Imperial table

MBL [Kips]	Diameter [Inches]	Mass in Air [lbs/ft]	Mass in Water [lbs/ft]	Maximum Length [ft]
1100	5 3/8	8.2	2.1	26893
1300	5 7/8	9.6	2.5	22944
1600	6 1/4	11.0	2.8	20006
1800	6 5/8	12.4	3.2	17832
2000	7	13.7	3.5	16075
2200	7 3/8	15.1	3.9	14602
2400	7 5/8	16.4	4.2	13436
2600	8	17.8	4.6	12390
2900	8 3/8	19.8	5.1	11145
3100	8 5/8	21.0	5.4	10482
3300	8 7/8	22.4	5.8	9829
3500	9 1/4	23.9	6.2	9216
3800	9 1/2	25.2	6.5	8745
4000	9 6/8	26.5	6.8	8306
4200	10	28.6	7.3	7720
4400	10 1/4	30.0	7.7	7357
4600	10 1/2	31.2	8.0	7056
4800	10 5/8	32.6	8.4	6765
5000	10 7/8	34.1	8.8	6471
5300	11 1/8	35.4	9.1	6226
5500	11 3/8	37.5	9.6	5880
5700	11 5/8	38.9	10.0	5667
6000	11 3/4	40.3	10.4	5477
6200	12	41.6	10.7	5300
6400	12 1/4	43.0	11.1	5127
6600	12 3/8	44.4	11.4	4971

Aceton group Ltd., engineering, mooring, and foundations.

Section 2 Contents

Chain



Chain Proof / Breaking Load Tables

Marine Chain (U2, U3) & Rig Chain (ORQ, ORQ +20%)

Mooring Chain (R3, R3S, R4, R4S, R5)



Common Link

Common Studless Link

Enlarged Link

End Link

5 Common Link Length

3 Link Adapter

5 Link Adapter

Chain / Mechanical Properties



Brazil

Egypt

Norway

Singapore

United Kingdom

United States

West Africa

Section 2

Chain Proof / Breaking Load Tables



Mooring Chain - 136 to 165mm

All loads are in Kn.

Chain (mm)	R3 Studless Proof	R3 Studlink Proof	R3 Break	R3S Studless Proof	R3S Studlink Proof	R3S Break	R4 Studless Proof	R4 Studlink Proof
136	9556	9556	13661	10659	11027	15253	11762	13232
137	9674	9674	13829	10790	11162	15441	11906	13395
138	9792	9792	13997	10922	11298	15629	12052	13558
139	9910	9910	14167	11054	11435	15818	12197	13722
140	10029	10029	14336	11186	11572	16008	12343	13886
141	10148	10148	14506	11319	11709	16198	12490	14051
142	10267	10267	14677	11452	11847	16388	12637	14216
143	10387	10387	14848	11585	11985	16579	12784	14382
144	10507	10507	15019	11719	12123	16770	12931	14548
145	10627	10627	15191	11853	12262	16962	13079	14714
146	10747	10747	15363	11987	12401	17154	13228	14881
147	10868	10868	15536	12122	12540	17347	13376	15048
148	10989	10989	15709	12257	12680	17540	13525	15216
149	11110	11110	15882	12392	12820	17734	13674	15384
150	11232	11232	16056	12528	12960	17928	13824	15552
151	11354	11354	16230	12664	13101	18122	13974	15721
152	11476	11476	16405	12800	13241	18317	14124	15890
153	11598	11598	16579	12936	13382	18512	14275	16059
154	11721	11721	16755	13073	13524	18708	14425	16229
155	11843	11843	16930	13210	13665	18904	14576	16399
156	11966	11966	17106	13347	13807	19100	14728	16569
157	12089	12089	17282	13484	13949	19297	14879	16739
158	12213	12213	17458	13622	14092	19493	15031	16910
159	12336	12336	17635	13760	14234	19691	15183	17081
160	12460	12460	17811	13898	14377	19888	15335	17252
161	12584	12584	17989	14036	14520	20086	15488	17424
162	12708	12708	18166	14174	14663	20284	15641	17596
163	12832	12832	18343	14313	14806	20482	15793	17768
164	12957	12957	18521	14452	14950	20681	15947	17940
165	13081	13081	18699	14590	15094	20879	16100	18112

Brazil

Egypt

Norway

Singapore

United Kingdom

United States

West Africa